Model Selection and Smoothing in Generalized Nonparametric Regression

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Helwig (2024) JCGS paper https://doi.org/10.1080/10618600.2024.2362232

Helwig (2024) **grpnet** R package https://cran.r-project.org/package=grpnet

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Multiple and Generalized Nonparametric Regression

Suppose that Y that follows an exponential family distribution.

Given predictors $(X_1, \ldots, X_p)$, consider a MGNR model of the form

$$g(\mu) = f(X_1, \ldots, X_p)$$

where

- μ is the (conditional) expectation of Y (given $X_1, \ldots, X_p$)
- $g(\cdot)$ is a user-specified link function (monotonic and invertible)
- $f(\cdot)$ is an unknown "smooth" function of $q \leq p$ predictors

Goal: estimate the unknown function $f(\cdot)$ from training data.

Challenge: Model Structure is Completely Unknown

Variable Selection Problem: which predictors should be included?

- Not all of the candidate predictors may be needed
- Want to determine which predictors are relevant (active)

Term Selection Problem: how should variables be included?

- Do the predictors combine in an additive fashion? Or interactive?
- Want to learn which main and/or interaction effects matter

Smoothing Problem: how smooth should each term be?

- Predictors may (and likely will) enter model in nonlinear fashion
- Want to avoid under-fitting (bias) and over-fitting (variability)

Three-Pronged Approach

Step 1: The Theory (Tensor Product Reproducing Kernels)

- Existing tensor product RKHS theory is not ideal for selection
- Develop improved representer theorem for tensor products

Step 2: The Algorithm (Adaptively Bounded Gradient Descent)

- Existing algorithms are not easily adaptable (e.g., to different g)
- Develop versatile group elastic net algorithm for GLMs/GAMs

Step 3: The Practice (Smoothing and Subspace Selection)

- Existing implementations only consider additive models
- Develop intuitive software for selecting main and interaction effects

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Generalized Linear Model

Consider a generalized linear model (GLM) of the form

$$g(\mu) = \boldsymbol{X}^\top \boldsymbol{\beta} \tag{1}$$

where

- $g(\cdot)$ is an invertible link function
- $\mu = E(Y|\boldsymbol{X})$ is the conditional expectation of Y given $\boldsymbol{X}$
- $\boldsymbol{X} = (X_1, \ldots, X_p)^\top \in \mathbb{R}^p$ is the predictor vector
- $\boldsymbol{\beta} \in \mathbb{R}^p$ is the unknown coefficient vector

The maximum likelihood (ML) estimates of the coefficients are

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta} \in \mathbb{R}^p} L(\boldsymbol{\beta}|\mathbf{D})$$

where

- $L(\boldsymbol{\beta}|\mathbf{D})$ denotes the negative of the log-likelihood function
- $\mathbf{D} = \{\mathbf{y}, \mathbf{X}\}$ is the observed data

Ridge, LASSO, and Elastic Net

Consider finding the coefficients that satisfy

$$\hat{\boldsymbol{\beta}}^*_{\lambda,\alpha} = \arg\min_{\boldsymbol{\beta} \in \mathbb{R}^p} \left\{ \frac{1}{n} L(\boldsymbol{\beta} | \mathbf{D}) + \lambda P^*_\alpha(\boldsymbol{\beta}) \right\}$$

where $\lambda \geq 0$ controls the overall influence of the penalty

$$P^*_\alpha(\boldsymbol{\beta}) = \alpha \sum_{j=1}^p |\beta_j| + \frac{1}{2}(1-\alpha) \sum_{j=1}^p \beta_j^2 \tag{2}$$

and $\alpha \in [0, 1]$ controls the weight of the lasso and ridge penalties.

- $\alpha = 0$ corresponds to the ridge (Hoerl & Kennard, 1970)
- $\alpha = 1$ corresponds to the lasso (Tibshirani, 1996)
- $\alpha \in (0, 1)$ corresponds to the elastic net (Zou & Hastie, 2005)

Regularization with Grouped Predictors

Multiple β_js may be needed to represent a single model term:

- categorical predictors, polynomial effects, regression splines
- interactions involving any of the above

Problem: the penalty in Equation (2) selects β_js individually (if $\alpha > 0$).

Solution(ish): Yuan & Lin (2006) proposed the group lasso penalty

$$P(\boldsymbol{\beta}) = \sum_{k=1}^{K} \omega_k \|\boldsymbol{\beta}_k\| \tag{3}$$

where

- $\omega_k \geq 0$ weights the k-th group's penalty (typically $\omega_k = \sqrt{p_k}$)
- $\|\boldsymbol{\beta}_k\| = (\boldsymbol{\beta}_k^\top \boldsymbol{\beta}_k)^{1/2}$ is the Euclidean norm of $\boldsymbol{\beta}_k$

Group Elastic Net Regularized GLM

I consider a groupwise extension of the classic elastic net penalty:

$$\hat{\boldsymbol{\beta}}_{\lambda,\alpha} = \underset{\boldsymbol{\beta} \in \mathbb{R}^p}{\arg\min} \left\{ \frac{1}{n} L(\boldsymbol{\beta} | \mathbf{D}) + \lambda P_\alpha(\boldsymbol{\beta}) \right\} \tag{4}$$

where the group elastic net penalty function has the form

$$P_\alpha(\boldsymbol{\beta}) = \sum_{k=1}^{K} \omega_k \left(\alpha \|\boldsymbol{\beta}_k\| + \frac{1}{2}(1-\alpha)\|\boldsymbol{\beta}_k\|^2 \right) \tag{5}$$

which is a natural groupwise extension of Equation (2).

Exponential Family Form

The response variable Y follows an exponential family distribution:

$$f_Y(y|\theta, \phi) = \exp\left\{ \frac{y\theta - b(\theta)}{a(\phi)} + c(y, \phi) \right\}$$

where

- θ is the canonical parameter
- ϕ is the dispersion parameter
- $a(\cdot)$, $b(\cdot)$, and $c(\cdot)$ are known functions

It is well-known that

$$E(Y) = \mu = b'(\theta)$$

$$\mathrm{Var}(Y) = \sigma^2 = b''(\theta)a(\phi)$$

where

- $b'(\cdot) = \mathrm{d}b(\theta)/\mathrm{d}\theta$ and $b''(\cdot) = \mathrm{d}^2 b(\theta)/\mathrm{d}\theta^2$
- $b''(\theta) = V(\theta)$ is referred to as the variance function

Likelihood Function

Given an independent sample of n observations, the log-likelihood function has the form

$$l(\boldsymbol{\beta}|\mathbf{D}) = \sum_{i=1}^{n} \left(\frac{y_i \theta_i - b(\theta_i)}{\phi / w_i} + c(y_i, \phi) \right)$$

where

- y_i is the i-th observation's response
- $\theta_i = \tilde{g}(g^{-1}(\eta_i))$ is the i-th observation's canonical parameter
- $\tilde{g}(\cdot)$ is the canonical link and $g(\cdot)$ is the chosen link
- $\eta_i = \mathbf{x}_i^\top \boldsymbol{\beta}$ is the i-th observation's linear predictor
- $\mathbf{x}_i = (x_{i1}, \ldots, x_{ip})^\top$ is the i-th observation's predictor vector
- $w_i > 0$ is the i-th observation's known weight

Note: this assumes that $a(\phi) = \phi / w$ for some known $w > 0$.

Penalized Likelihood Estimation

Dropping $c(y_i, \phi)$ and multiplying by $-\frac{1}{n}$ produces the loss function

$$L(\boldsymbol\beta|\mathbf{D}) = -\frac{1}{n}\sum_{i=1}^{n} w_i \left(y_i\theta_i - b(\theta_i)\right)$$

which sets $\phi = 1$ (WLOG) for the purposes of estimating $\boldsymbol\beta$.

The desired penalized likelihood estimates satisfy

$$\hat{\boldsymbol\beta}_{\lambda,\alpha} = \underset{\boldsymbol\beta \in \mathbb{R}^p}{\arg\min} \left\{ -\frac{1}{n}\sum_{i=1}^{n} w_i \left(y_i\theta_i - b(\theta_i)\right) + \lambda P_\alpha(\boldsymbol\beta) \right\}$$

where $\theta_i = \tilde{g}(g^{-1}(\mathbf{x}_i^\top \boldsymbol\beta))$ implicitly depends on the coefficient vector $\boldsymbol\beta$

Groupwise Penalized Likelihood Estimation

Focus on cyclical algorithms that iteratively estimate $\boldsymbol{\beta}_k$ given others.

I consider iteratively solving (for $k = 1, \ldots, K$) the problem

$$\hat{\boldsymbol{\beta}}_{\lambda,\alpha}^{(k)} = \underset{\boldsymbol{\beta}_k \in \mathbb{R}^{p_k}}{\arg\min} \left\{ L(\boldsymbol{\beta}|\mathbf{D}) + \lambda P_\alpha^{(k)}(\boldsymbol{\beta}_k) \right\} \tag{6}$$

where

$$P_\alpha^{(k)}(\boldsymbol{\beta}_k) = \omega_k \left(\alpha \|\boldsymbol{\beta}_k\| + \frac{1}{2}(1-\alpha)\|\boldsymbol{\beta}_k\|^2 \right)$$

is the k-th group's penalty contribution.

Problem: there doesn't exist a closed-form solution for general cases.

Gradient Vector

The negative gradient of $L(\boldsymbol{\beta}|\mathbf{D})$ with respect to $\boldsymbol{\beta}_k$ is

$$\mathbf{g}_k = -\left(\frac{\partial L(\boldsymbol{\beta}|\mathbf{D})}{\partial \boldsymbol{\beta}_k}\right) = \frac{1}{n}\sum_{i=1}^{n} \frac{w_i}{d_i v_i}(y_i - \mu_i)\mathbf{x}_{ik} \tag{7}$$

where

- $v_i = b''(\theta_i)$ is the i-th observation's variance function realization
- $d_i = \left(\frac{\mathrm{d}\eta}{\mathrm{d}\mu}\Big|_{\mu=\mu_i}\right)$ is the derivative $g'(\mu)$ evaluated at $\mu_i = g^{-1}(\mathbf{x}_i^\top \boldsymbol{\beta})$

Table 1 gives the v_i for many commonly used response distributions. Table 2 gives the d_i for many commonly used link functions.

Variance Functions

Table 1: Exponential family parameterizations and variance functions.

Family	Parameterization	ϕ	v_i
Gaussian	$N(\mu, \sigma^2)$	σ^2	1
Binomial	$B(\mu)$	1	$\mu_i(1 - \mu_i)$
Multinomial	$M(\mu)$	1	$\mu_{i\ell}(1 - \mu_{i\ell})$
Poisson	$P(\mu)$	1	μ_i
Negative Binomial	$NB(\mu, \theta)$	1	$\mu_i + \mu_i^2/\theta$
Gamma	$G(\mu, \nu)$	$1/\nu$	μ_i^2
Inverse Gaussian	$IG(\mu, \sigma^2)$	σ^2	μ_i^3

Link Functions

Table 2: Link functions $g(\cdot)$, their inverses $g^{-1}(\cdot)$, and their derivatives $g'(\cdot)$.

Link	$\eta = g(\mu)$	$\mu = g^{-1}(\eta)$	$d_i = g'(\mu_i)$
Identity	μ	η	1
Logit	$\log(\frac{\mu}{1-\mu})$	$\frac{\exp(\eta)}{1+\exp(\eta)}$	$\frac{1}{\mu_i(1-\mu_i)}$
Multi-Logit	N/A	$\frac{\exp(\eta_\ell)}{\sum_{l=1}^m \exp(\eta_l)}$	$\frac{1}{\mu_{i\ell}(1-\mu_{i\ell})}$
Log	$\log(\mu)$	$\exp(\eta)$	$1/\mu_i$
Inverse	$1/\mu$	$1/\eta$	$-1/\mu_i^2$
Inverse-Square	$1/\mu^2$	$1/\eta^{1/2}$	$-2/\mu_i^3$

Penalty Derivatives

Assuming $\boldsymbol{\beta}_k \neq \mathbf{0}_{p_k}$, the gradient of the group lasso penalty $P_1(\boldsymbol{\beta})$ is

$$\mathbf{q}_{1k} = \frac{\partial P_1(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}_k} = \frac{\omega_k}{\|\boldsymbol{\beta}_k\|}\boldsymbol{\beta}_k$$

and note that $\mathbf{q}_{1k}$ is not defined if $\boldsymbol{\beta}_k = \mathbf{0}_{p_k}$.

The gradient of the group ridge penalty $P_2(\boldsymbol{\beta})$ is

$$\mathbf{q}_{2k} = \frac{\partial P_2(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}_k^\top} = \omega_k \boldsymbol{\beta}_k$$

for all $\boldsymbol{\beta}_k \in \mathbb{R}^{p_k}$.

Optimality Conditions

A necessary condition for the vector $\boldsymbol{\beta}^\top = (\boldsymbol{\beta}_1^\top, \ldots, \boldsymbol{\beta}_K^\top)$ to be a solution to Equation (4) is that

$$\begin{aligned} -\mathbf{g}_k + \lambda\left(\alpha \mathbf{q}_{1k} + (1-\alpha)\mathbf{q}_{2k}\right) = \mathbf{0}_{p_k} \qquad & \forall \boldsymbol{\beta}_k \neq \mathbf{0}_{p_k} \\ \|\mathbf{g}_k\| \leq \alpha\lambda\omega_k \qquad & \forall \boldsymbol{\beta}_k = \mathbf{0}_{p_k} \end{aligned} \tag{8}$$

which follows directly from the Karush-Kuhn-Tucker (KKT) conditions (Karush, 1939; Kuhn & Tucker, 1951; Yang & Zou, 2015).

Active groups' coefficients satisfy standard gradient conditions
Inactive groups' coefficients satisfy subgradient conditions

Fisher Information and Unpenalized Solution

The Fisher information matrix for the loss function L has the form

$$\mathbf{H}_k = E\left(\frac{\partial^2 L(\boldsymbol\beta|\mathbf{D})}{\partial\boldsymbol\beta_k \partial\boldsymbol\beta_k^\top}\right) = \frac{1}{n}\sum_{i=1}^n \frac{w_i}{d_i^2 v_i}\mathbf{x}_{ik}\mathbf{x}_{ik}^\top \tag{9}$$

which is positive definite if columns of $\mathbf{X}_k$ are linearly independent.

When $\lambda = 0$, the MLEs can be obtained via cyclic Fisher scoring:

$$\boldsymbol\beta_k^{(t+1)} = \boldsymbol\beta_k^{(t)} + \left(\mathbf{H}_k^{(t)}\right)^{-1}\mathbf{g}_k^{(t)} \tag{10}$$

where

- $\mathbf{g}_k^{(t)}$ and $\mathbf{H}_k^{(t)}$ are gradient and information evaluated at $\boldsymbol\beta_k^{(t)}$
- $t = 0, 1, 2, \ldots$ denotes the iteration (step) counter

Local Quadratic Approximation

$\boldsymbol{\beta}_k^* = \left(\tilde{\boldsymbol{\beta}}_1^\top, \ldots, \tilde{\boldsymbol{\beta}}_K^\top\right)$ with $\tilde{\boldsymbol{\beta}}_j = \boldsymbol{\beta}_j$ for $j \in \{1, \ldots, K\} \backslash \{k\}$ and $\tilde{\boldsymbol{\beta}}_k \neq \boldsymbol{\beta}_k$.

Local quadratic approximation of $L(\boldsymbol{\beta}|\mathbf{D})$ near $\boldsymbol{\beta}_k^*$ has the form

$$\begin{aligned} L(\boldsymbol{\beta}|\mathbf{D}) &\approx L_k(\boldsymbol{\beta}_k|\mathbf{D}, \boldsymbol{\beta}_k^*) \\ &= L(\boldsymbol{\beta}_k^*|\mathbf{D}) - \left(\boldsymbol{\beta}_k - \tilde{\boldsymbol{\beta}}_k\right)^\top \tilde{\mathbf{g}}_k + \frac{1}{2}\left(\boldsymbol{\beta}_k - \tilde{\boldsymbol{\beta}}_k\right)^\top \tilde{\mathbf{H}}_k \left(\boldsymbol{\beta}_k - \tilde{\boldsymbol{\beta}}_k\right) \end{aligned}$$

where $\tilde{\mathbf{g}}_k$ and $\tilde{\mathbf{H}}_k$ denote gradient and information evaluated at $\tilde{\boldsymbol{\beta}}_k$.

Setting the derivative to zero to solve for $\boldsymbol{\beta}_k$ gives

$$\boldsymbol{\beta}_k = \tilde{\boldsymbol{\beta}}_k + \tilde{\mathbf{H}}_k^{-1}\tilde{\mathbf{g}}_k$$

which is Equation (10) with $\tilde{\boldsymbol{\beta}}_k$ serving as the current estimate

Majorized Loss Function and Unpenalized Update

Consider minimizing the modified loss function

$$L_k^*(\boldsymbol{\beta}_k|\mathbf{D}, \boldsymbol{\beta}_k^*) = L(\boldsymbol{\beta}_k^*|\mathbf{D}) - (\boldsymbol{\beta}_k - \tilde{\boldsymbol{\beta}}_k)^\top \tilde{\mathbf{g}}_k + \frac{\gamma_k \delta}{2} \|\boldsymbol{\beta}_k - \tilde{\boldsymbol{\beta}}_k\|^2$$

where

- γ_k is the largest eigenvalue of $\frac{1}{n}\mathbf{X}_k^\top \mathbf{W} \mathbf{X}_k$
- $\delta = \max_i \left(1/d_i^2 v_i\right) = \min_i(d_i^2 v_i)$ (fixed or adaptively updated)

Lemma: L_k^* majorizes the quadratic approximation function L_k, i.e., $L_k(\boldsymbol{\beta}_k|\mathbf{D}, \boldsymbol{\beta}_k^*) \leq L_k^*(\boldsymbol{\beta}_k|\mathbf{D}, \boldsymbol{\beta}_k^*)$ for all $\boldsymbol{\beta}_k \in \mathbb{R}^{p_k}$.

The modified (majorized) Fisher scoring update

$$\boldsymbol{\beta}_k^{(t+1)} = \boldsymbol{\beta}_k^{(t)} + \left(\gamma_k \delta_{(t)}\right)^{-1} \mathbf{g}_k^{(t)} \qquad (11)$$

where $\delta_{(t)}$ is defined using the current coefficient estimates $\boldsymbol{\beta}_k^{(t)}$.

Adaptively Bounded Gradient Descent Update

Consider finding the $\boldsymbol{\beta}_k$ that satisfies

$$\underset{\boldsymbol{\beta}_k \in \mathbb{R}^{p_k}}{\arg\min} \left\{ L_k^*(\boldsymbol{\beta}_k | \mathbf{D}, \boldsymbol{\beta}_k^*) + \lambda P_\alpha^{(k)}(\boldsymbol{\beta}_k) \right\} \tag{12}$$

which is Equation (6) with L_k^* replacing L_k.

Theorem: The solution to Equation (6) can be obtained by cyclically solving Equation (12) using updates of the form

$$\boldsymbol{\beta}_k^{(t+1)} = \underbrace{\frac{1}{1 + (1-\alpha)\lambda\omega_k(\gamma_k\delta_{(t)})^{-1}}}_{L2} \underbrace{\left(1 - \frac{\alpha\lambda\omega_k(\gamma_k\delta_{(t)})^{-1}}{\left\|\mathbf{b}_k^{(t+1)}\right\|}\right)_+}_{L1} \underbrace{\mathbf{b}_k^{(t+1)}}_{ML}$$

where

- $\mathbf{b}_k^{(t+1)} = \boldsymbol{\beta}_k^{(t)} + \left(\gamma_k\delta_{(t)}\right)^{-1} \mathbf{g}_k^{(t)}$ is the unpenalized update
- $\delta_{(t)} = \min_i(d_i^2 v_i)$ where d_i and v_i are evaluated using $\boldsymbol{\beta}_k^{(t)}$

ABGD Algorithm for GLMs with Group Elastic Net

Algorithm 1 Adaptively bounded gradient descent algorithm for generalized linear models with group elastic net penalties.

Calculate $\gamma_k = \text{maxeig}\left(\frac{1}{n}\sum_{i=1}^n w_i \mathbf{x}_{ik}\mathbf{x}_{ik}^\top\right)$ for $k = 1, \ldots, K$
Initialize $\boldsymbol{\beta}_k = \mathbf{0}_{p_k}$ for all $k = 1, \ldots, K$
while $\max_j \frac{|\beta_j - \beta_{j(\text{old})}|}{1+|\beta_j|} > \epsilon$ **do**
    $\boldsymbol{\beta}_{\text{old}} \leftarrow \boldsymbol{\beta}$
    $\delta = \min_i(d_i^2 v_i) = \max_i(1/d_i^2 v_i)$
    for $k = 1, \ldots, K$ **do**
        $\mathbf{g}_k = \frac{1}{n}\sum_{i=1}^n \frac{w_i}{d_i v_i}(y_i - \mu_i)\mathbf{x}_{ik}$
        $\mathbf{b}_k = \boldsymbol{\beta}_k + (\gamma_k \delta)^{-1}\mathbf{g}_k$
        $\boldsymbol{\beta}_k = \frac{1}{1+(1-\alpha)\lambda\omega_k(\gamma_k\delta)^{-1}}\left(1 - \frac{\alpha\lambda\omega_k(\gamma_k\delta)^{-1}}{\|\mathbf{b}_k\|}\right)_+ \mathbf{b}_k$
    end for
end while

Relation to Existing Algorithms

Closely related to the update proposed by Yang & Zou (2015):

- $\alpha = 1$ and $\delta_{(t)} = \delta_* \; \forall t$ (i.e., no separation of γ_k and δ)
- Only consider Gaussian, binomial, and two SVM loss functions

Also related to the update proposed by Breheny & Huang (2015):

- $\alpha = 1$, $\gamma_k = 1$ (orthogonalized), and $\delta_{(t)} = \delta_* \; \forall t$
- Only consider Gaussian, binomial, Poisson, and Cox PH

Neither separates $\mathbf{H}_k$ into fixed design contribution (bounded by γ_k) and adaptive mean-dependent contribution (bounded by δ).

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Non-Exhaustive List of R Packages for Group LASSO

There are a handful of options for fitting group lasso in R:

- **gglasso** package (Yang et al., 2020)
- **grplasso** package (Meier, 2020)
- **grpnet** package (Helwig, 2024)
- **grpreg** package (Breheny, 2021)

Note that **grplasso** isn't computationally competitive (no majorization)

Other packages all use majorization—but with practical differences!

Scope of Packages: Loss Functions

Table 3: Loss functions available in each package.

$L(\boldsymbol{\beta}\|\mathbf{D})$	**gglasso**	**grplasso**	**grpnet**	**grpreg**
Gaussian	✓	✓	✓	✓
Binomial	✓	✓	✓	✓
Poisson		✓	✓	✓
Multinomial			✓	
Negative Binomial			✓	
Gamma			✓	
Inverse Gaussian			✓	
Cox Proportional Hazard				✓
Squared SVM	✓			
Huberized SVM	✓			

Scope of Packages: Usability Features

Table 4: Usability features of each package.

Features	gglasso	grplasso	grpnet	grpreg
α and γ tuning			✓	
Parallelization			✓	
Default method	✓		✓	✓
Formula method		✓	✓	
Orthogonalization		✓	✓	✓
Standardization			✓	
Additive splines			✓	✓
Tensor product splines			✓	
MCP and SCAD			✓	✓

Some Improved Selection Penalties: SCAD and MCP

Table 5: Different L1 penalties and their derivatives used in **grpnet** package.

	Penalty Function $Q_\lambda(\theta)$	Penalty Derivative $\frac{d}{d\theta}Q_\lambda(\theta)$
LASSO	$\lambda\lvert\theta\rvert$	λ
MCP	$\begin{cases} \lambda\lvert\theta\rvert - \frac{\theta^2}{2\gamma} & \text{if } \lvert\theta\rvert \leq \gamma\lambda \\ \gamma\lambda^2/2 & \text{otherwise} \end{cases}$	$\begin{cases} \lambda - \frac{\lvert\theta\rvert}{\gamma} & \text{if } \lvert\theta\rvert \leq \gamma\lambda \\ 0 & \text{otherwise} \end{cases}$
SCAD	$\begin{cases} \lambda\lvert\theta\rvert & \text{if } \lvert\theta\rvert \leq \lambda \\ \frac{\gamma\lambda\lvert\theta\rvert - (\theta^2+\lambda^2)/2}{\gamma-1} & \text{if } \lambda < \lvert\theta\rvert \leq \gamma\lambda \\ (\gamma+1)\lambda^2/2 & \text{otherwise} \end{cases}$	$\begin{cases} \lambda & \text{if } \lvert\theta\rvert \leq \lambda \\ \frac{\gamma\lambda - \lvert\theta\rvert}{\gamma-1} & \text{if } \lambda < \lvert\theta\rvert \leq \gamma\lambda \\ 0 & \text{otherwise} \end{cases}$

Visualization of Penalty and Derivative Functions

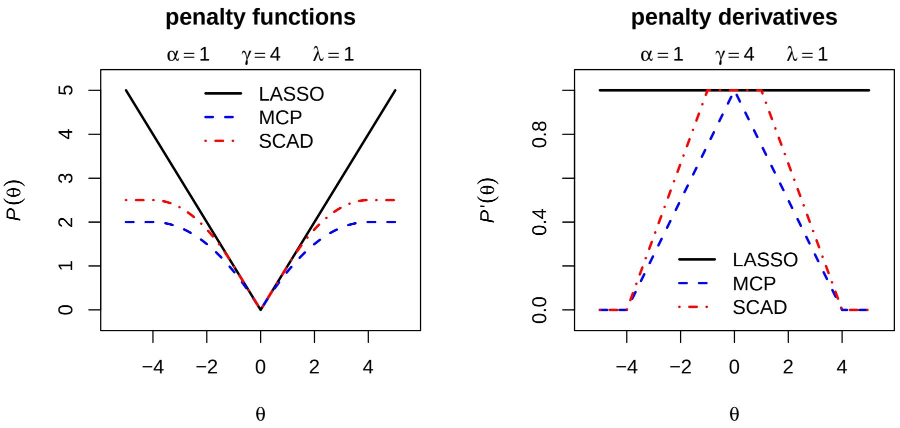

Figure 1: Different L1 penalties and their derivatives used in **grpnet** package.

Visualization of Shrinkage and Selection Operators

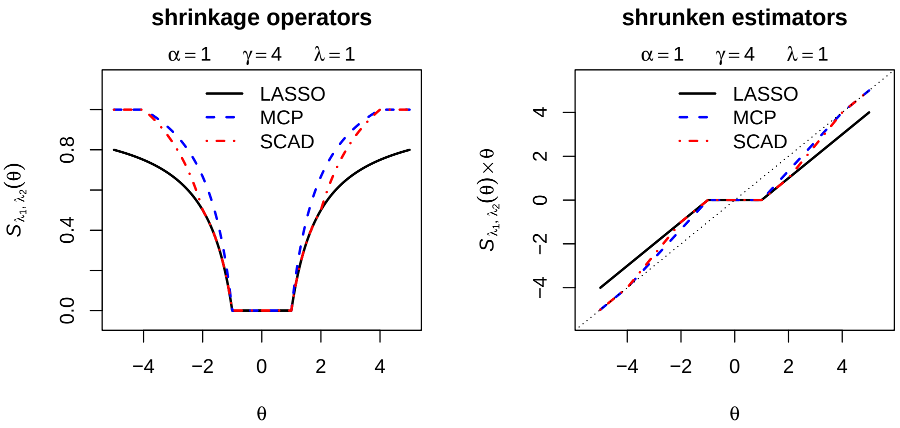

Figure 2: Shrinkage and selection effects of L1 penalties used in **grpnet** package.

Simulation A: Design and Analyses

Fully crossed simulation design with three factors:

1. response distribution (3 levels: Gaussian, Binomial, Poisson)
2. sample size (2 levels: $n \in \{100, 1000\}$)
3. number of predictors (10 levels: $p = 10 \times 2^m$ for $m \in \{1, \ldots, 10\}$)

Data generation and analysis details:

- group size was fixed at $p_k = 5$ for all $k = 1, \ldots, K$
- elements of $\mathbf{X}$ were i.i.d. $N(0, 1)$
- compared **grpnet**, **grpreg**, and **gglasso**
- calculate time to compute regularization path using 100 λs
- repeat entire procedure 10 times for each simulation cell

Simulation A: Timing Results ($n = 100$)

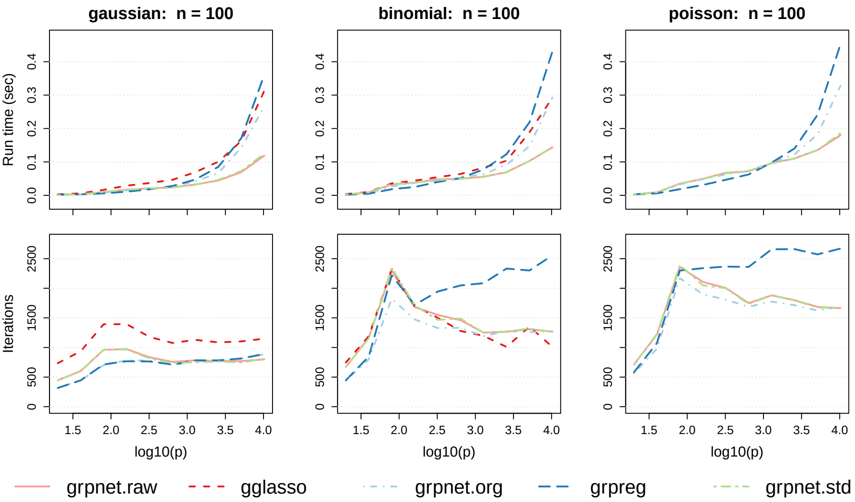

Figure 3: Amount of time to compute the regularization path with $n = 100$.

Simulation A: Timing Results ($n = 1000$)

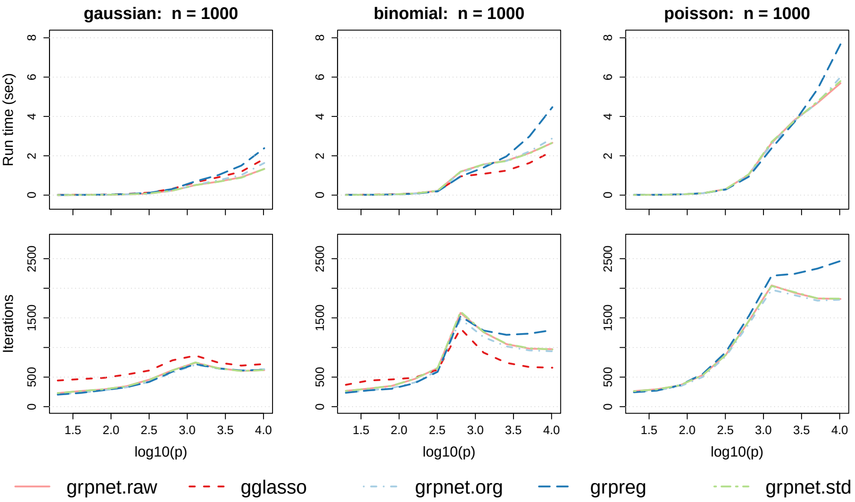

Figure 4: Amount of time to compute the regularization path with $n = 1000$.

Simulation B: Data Generating Model

Data Generating Mean Function Form:

$$\mu(X) = \alpha_1 f_1(X_1) + \alpha_2 f_2(X_2) + \alpha_3 f_3(X_1, X_2)$$

where

- f_1 and f_2 are the main effect functions
- f_3 is the interaction effect function
- $(\alpha_1, \alpha_2, \alpha_3)$ are varied to control size of effects
- 3 active terms: X_1, X_2, and $X_1 : X_2$

Independently generate data of the form

$$Y_i = \mu(X_i) + \epsilon_i$$

where X_{ij} are i.i.d. $U[0, 1]$ and $\epsilon_i \sim N(0, \sigma^2)$.

Simulation B: Component Functions

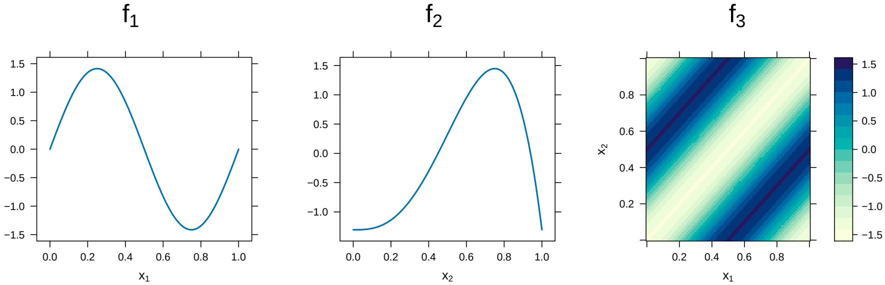

Figure 5: Component functions used in simulation. $f_1(x) \propto \sin(2\pi x)$ (sine function), $f_2(x) \propto \text{Beta}(x; 4, 2) - 1$ (beta density function), and $f_3(x_1, x_2) \propto -\cos(2\pi|x_1 - x_2|)$ (cosine function). All component functions were centered (to have mean zero) and scaled (to have unit variance) for the domain $\mathcal{X} = [0, 1] \times [0, 1]$.

Simulation B: Data Generating Functions

Consider three different mean functions:

(A) $\alpha_1 = \alpha_2 = \alpha_3 = 1$ (equal strength effects)

(B) $\alpha_1 = \alpha_2 = 1.2$ and $\alpha_3 = 0.6$ (weak interaction effect)

(C) $\alpha_1 = \alpha_2 = 0.6$ and $\alpha_3 = 1.8$ (weak main effects)

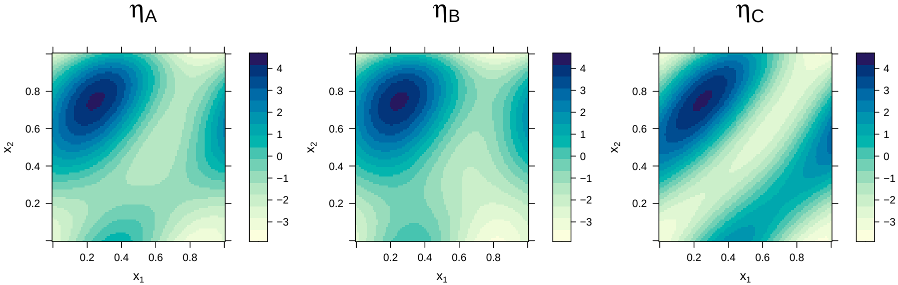

Figure 6: Mean functions formed via linear combinations of the component functions.

Simulation B: Design and Analyses

Fully crossed design that manipulates two factors:

- $n \in \{100, 250, 500, 750, 1000\}$
- $p \in \{10, 20\}$

Cubic spline basis with 5 df for each marginal

- Each main effect has 5 coefficients
- Each two-way interaction effect has 25 coefficients

Fit model with all p main effects and all $\binom{p}{2}$ two-way interactions:

p	$\binom{p}{2}$	K	# coefs
10	45	55	1175
20	190	210	4850

Simulation B: Model Recovery Results ($p = 10$)

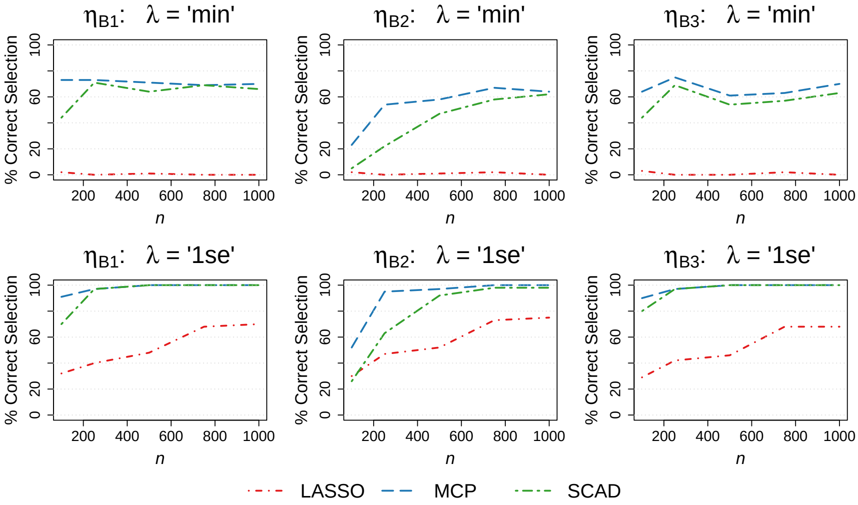

Figure 7: Percentage of time 10-fold CV selected the correct model when $p = 10$.

Simulation B: Model Recovery Results ($p = 20$)

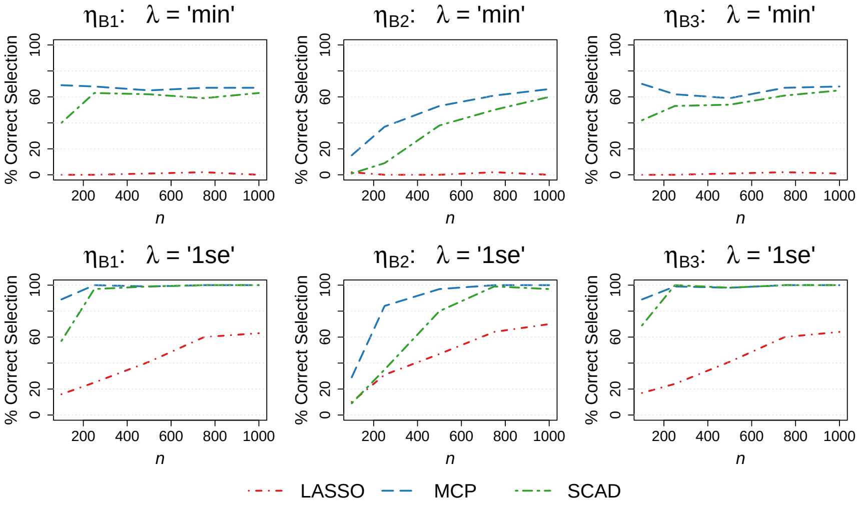

Figure 8: Percentage of time 10-fold CV selected the correct model when $p = 20$.

Real Dataset Details

I consider five different datasets from the UCI-MLR (Kelly et al., 2024)

Table 6: Features of the real datasets used in the examples.

Dataset	Response	Family	n	p	K
Wine Quality	quality	Gaussian	4898	55	11
Wisconsin Breast Cancer (Original)	Class	Binomial	683	80	9
Student Performance (Math)	failures	Poisson	395	89	32
Splice-Junction Gene Sequences	border	Multinomial	3190	227	60
Gene Expression Cancer RNA-Seq	labels	Multinomial	801	101320	20264

Note: K is the number of predictors (features), whereas $p \geq K$ is the number of columns of the design matrix after expanding:

- categorical predictors using dummy coding
- continuous predictors using B-spline basis

Real Data Analyses

The `bs()` function in **splines** package (R Core Team, 2024) used to expand each continuous predictor using a B-spline basis with `df = 5`.

For the first three examples, I compared **grpnet**, **grpreg**, & **gglasso** by

- randomly splitting data into training (80%) and testing (20%) sets
- applying 10-fold CV to tune λ using the training data
- evaluating the trained models' predictions using the testing data

For the two multinomial examples, I demonstrate the functionality of the `cv.grpnet()` function and corresponding S3 plotting method.

Real Data Results: Methods Comparison

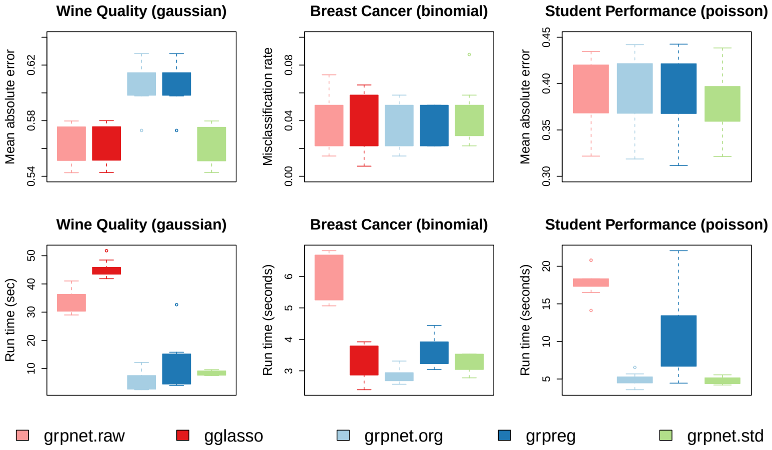

Figure 9: Prediction error for test data (top) and tuning time for training data (bottom).

Real Data Results: CV Error Curves

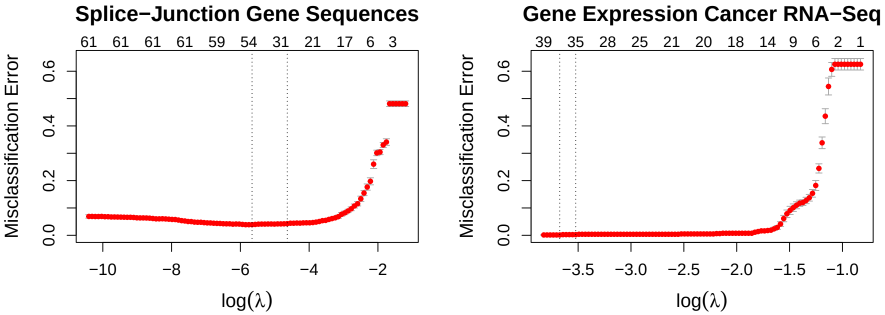

Figure 10: Cross-validation error curves (i.e., average misclassification rate by $\log(\lambda)$) with standard error bars for the multinomial examples. Within each panel, the vertical dotted lines show: the λ that minimizes the cross-validation error (left), and the largest λ that is within one standard error of the minimum (right). The numbers on the top axis show the number of non-zero groups as a function of $\log(\lambda)$.

Cancer-RNA Solution Print Out

Note: the optimal solution only has 39 groups (= 38 genes + intercept)

```
> cvmod.cancer

Call:   cv.grpnet.default(x = xbs, y = y, group = g, alpha = 1, parallel = TRUE,
    cluster = cl, family = "multinomial")

Measure:    Misclassification Error

     Alpha  Lambda Index Measure       SE nzGroup nzCoef    Df
min      1 0.02538    95 0.00125 0.001250      39    955 428.5
1se      1 0.02953    90 0.00250 0.001667      35    855 380.0
```

10-fold CV prediction accuracy is 99.875% (min) or 99.75% (1se)

Cancer-RNA Classification Performance Comparison

TABLE II. TRAINING AND TESTING SCORE ANALYSIS

Model	Training Score (%)	Testing Score (%)
LR	99.46	98.59
NB	98.75	99.17
RF	99.46	98.76
KNN	99.46	98.75
DT	98.75	97.51
Proposed	***99.64***	***99.59***

Note. Reproduction of Table II from Wahid & Banday (2023).

grpnet achieves 99.75% (training) and 99.20% (testing) only tuning λ

Cancer-RNA Biomarkers via Effect Size Plots

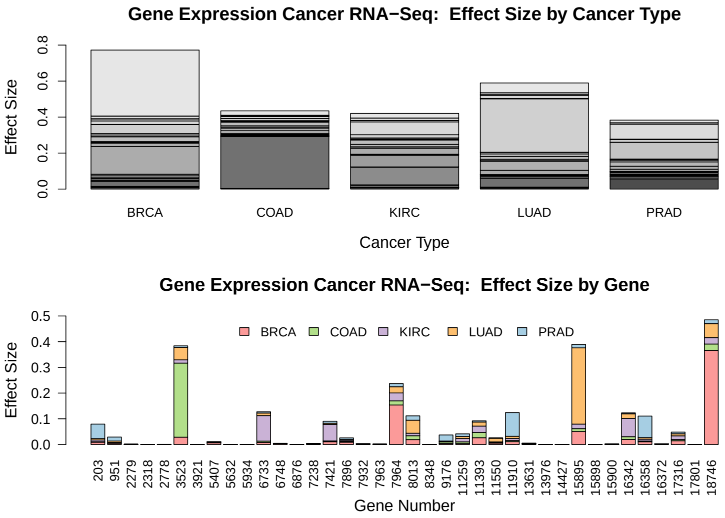

Figure 11: Effect sizes, i.e., $\xi_{k\ell} = n^{-1}\|\mathbf{C}_n \mathbf{X}_k \boldsymbol{\beta}_{k\ell}\|^2$, for the $K = 38$ non-zero genes.

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Bike Sharing: Dataset and Analyses

Bike sharing dataset from the UCI-MLR (Fanaee-T & Gama, 2013).

- $n = 17{,}379$ observations
- $p = 11$ predictors

Predicting number of bike rentals (Poisson) from various features:

- Temporal: year, month, season, hour of day, weekday, holiday
- Weather: temperature, humidity, precipitation, wind speed

Fit model with all main effects (11) and all two-way interactions (55)

- Use 5 knots for continuous main effects
- Use 25 knots for two-way interaction effects

Bike Sharing: Variable Importance Indices

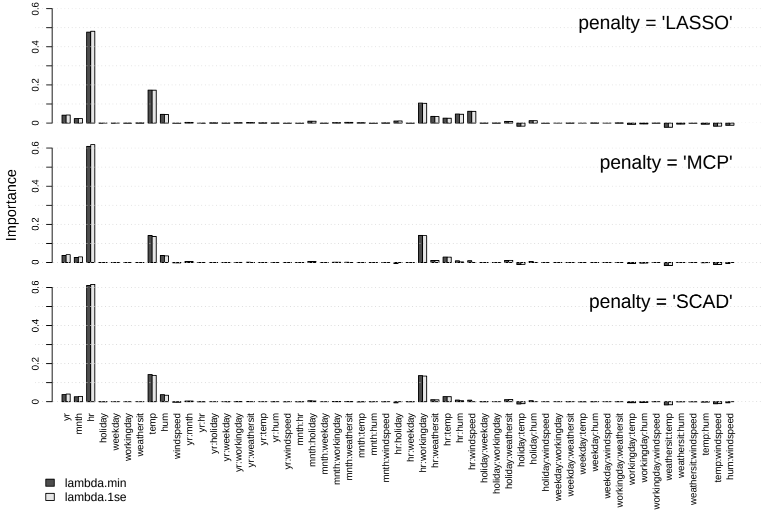

Figure 12: Variable importances indices for effects in bike sharing model.

Bike Sharing: Estimated Effects

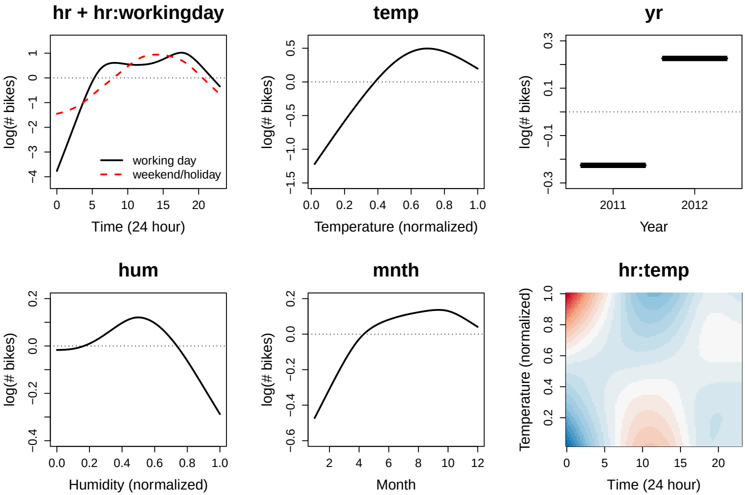

Figure 13: Estimated effects (linear predictors) for important terms using MCP.

Resting-State fMRI: Dataset and Analyses*

Data from Adolescent Brain Cognitive Development (ABCD) study

- $n = 7979$ children ages 9-10 years old
- 21 collection sites across the USA (including Minnesota)

Response: Attention Problems from Child Behavior Checklist (Poisson)

Predictors:

- 78 pairs of resting state network correlations (12 brain networks)
- Parent history of alcohol and drug use
- Collection site ID, Income, and Education
- Demographic information (age, sex, race)

*Joint work with Kelly Duffy

Resting-State fMRI: Functional Brain Networks

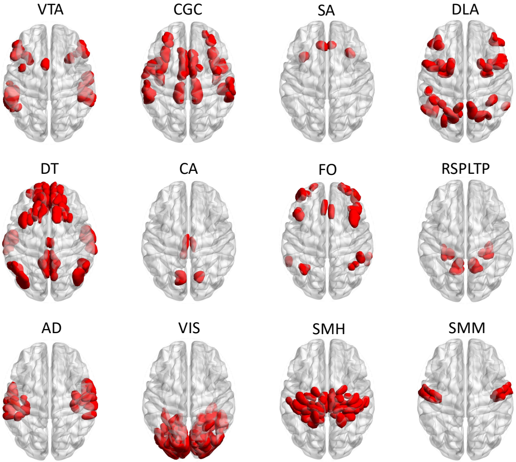

Figure 14: The 12 functional brain networks considered in this study.

Resting-State fMRI: Variable Importances Indices

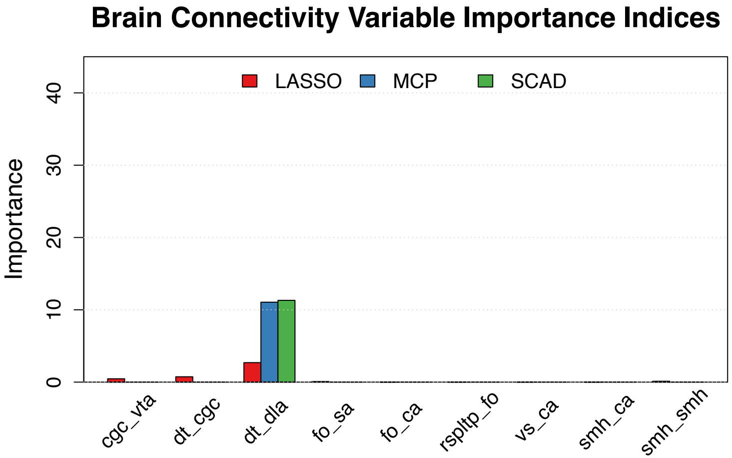

Figure 15: Variable importances indices for network effects in fMRI model.

Resting-State fMRI: Estimated Effect

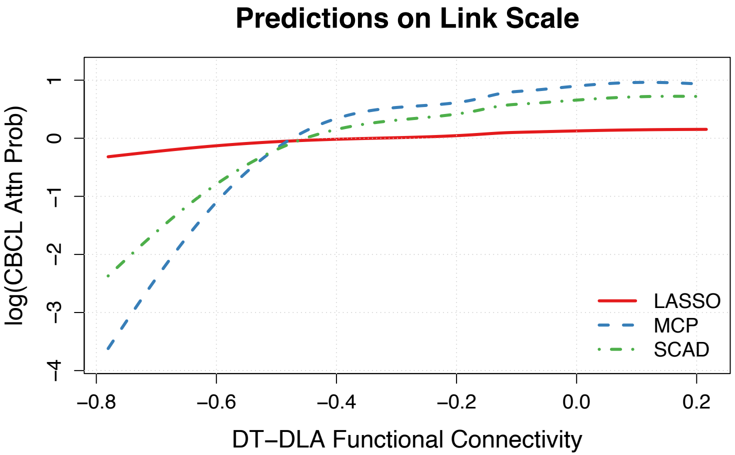

Figure 16: Effect for Default Mode (DT) and Dorsal Attention (DLA) connection.

Scientific Serendipity: Penguin from Left Field

Recently Trevor Hastie and his PhD student James Yang published:

- **adelie** Python package (with R port)
- arXiv paper about group elastic net

Timeline of events

- 2023-07-13 **grpnet** on CRAN
- 2023-08-20 Helwig JCGS paper submitted
- 2024-05-14 Yang and Hastie arXiv paper submitted
- 2024-05-25 Helwig JCGS paper accepted
- 2024-06-17 **adelie** on CRAN

Alternative Approach to the Problem

Yang and Hastie take a very different approach to problem:

- Solve weighted Gaussian problem using efficient Newton algorithm
- Use adaptive bisection method to find good initial point

Claim their method is 3-10 times more efficient than other methods:

- 3 times faster than my R package
- 10 times faster than others

Timing difference is primarily achieved through early exists of λ path.

Comparison of the Packages: Classic LASSO ($p_k = 1$)

```
> # example from ?adelie::grpnet
> set.seed(0)
> n <- 100
> p <- 200
> X <- matrix(rnorm(n * p), n, p)
> y <- X[,1] * rnorm(1) + rnorm(n)

> # adelie
> system.time({
+     fit1 <- adelie::grpnet(X, adelie::glm.gaussian(y), early_exit = FALSE)
+ })
   user  system elapsed
   0.01    0.00    0.01

> # grpnet
> system.time({
+     fit2 <- grpnet::grpnet(X, y)
+ })
   user  system elapsed
  0.020   0.001   0.021
```

Comparison of the Packages: Grouped LASSO ($p_k = 5$)

```
> # group size = 5
> grp <- rep(1:(p/5), each = 5)

> # adelie
> system.time({
+     fit1 <- adelie::grpnet(X, adelie::glm.gaussian(y), groups = grp)
+ })
Error: adelie_core: screen_beta has smaller length than screen_set.
It is likely screen_beta has been initialized incorrectly.

> # grpnet
> system.time({
+     fit2 <- grpnet::grpnet(X, y, group = grp)
+ })
   user  system elapsed
  0.015   0.000   0.015
```

Comparison of the Packages: Grouped LASSO ($p_k = 10$)

```
> # group size = 10
> grp <- rep(1:(p/10), each = 10)

> # adelie
> system.time({
+     fit1 <- adelie::grpnet(X, adelie::glm.gaussian(y), groups = grp)
+ })
Error: adelie_core: screen_beta has smaller length than screen_set.
It is likely screen_beta has been initialized incorrectly.

> # grpnet
> system.time({
+     fit2 <- grpnet::grpnet(X, y, group = grp)
+ })
   user  system elapsed
  0.009   0.000   0.009
```

Future Directions

Some short-term extensions:

- Early Stopping: λ sequence could be stopped early
- Adaptive Weights: ω_ks could adaptively estimated
- Multivariate Knots: need better strategy (multivariate quantiles?)

Some mid-range extensions

- Ordinal Responses: common in psychology and health sciences
- Multivariate Responses: ordinal and continuous mixtures
- Uncertainty Quantification: bootstrap or other sampling?

Some longer-term extensions

- ABGD update within dimension reduction methods
- Extensions to multivariate quantile regression problems

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